THORACIC WALL

Department of Anatomy

Thorax (1)

First part

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Lecture Points

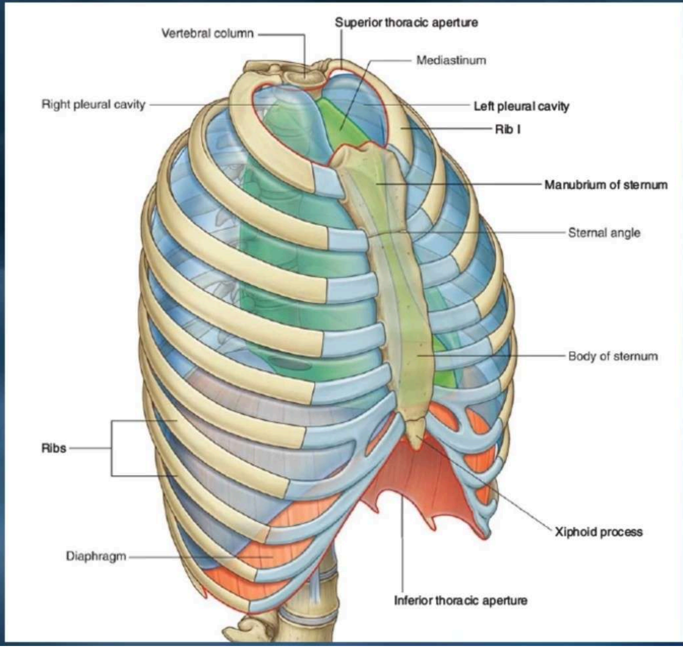
- Anatomy of the Thoracic wall and Thoracic Cavity.
- The superior thoracic aperture consists.
- The suprapleural membrane
- The bone Structure of the Thoracic Cage.
  - Sternum.
  - The sternal angle.
  - Ribs (Typical & Atypical).
  - Thoracic Vertebra (Typical & Atypical).
  - Joints (articulations) of the thorax
- Inferior Boundary (outlet) of the thorax.
- The diaphragm.
  - ✓ Physiological action of diaphragm.
  - ✓ Three Large Openings of Diaphragm.
  - ✓ arterial supply of the Diaphragm.
- The muscles of the thorax wall
- The arterial supply to the thoracic wall
- The venous drainage of the thoracic wall.
- lymphatic drainage of the thorax wall.
- Intercostal nerve (Typical & Atypical).
- Segmental innervation (dermatomes) of thoracic wall.
- herpes zoster the Spinal Ganglia
- Vertebral levels of sternum and transverse thoracic plane.
- Surface Anatomy of the Thoracic Wall Skeleton.

Anatomy of the Chest

1) Thoracic wall

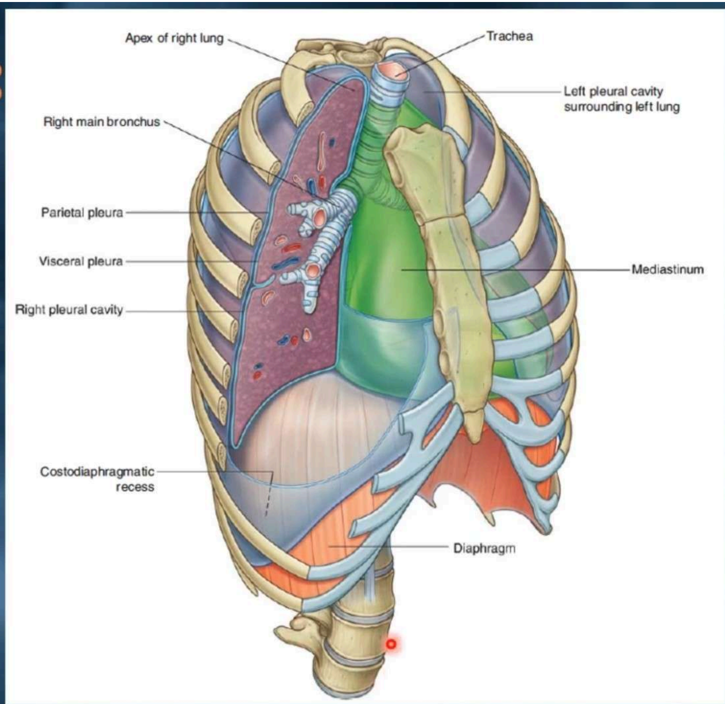
Formed by :

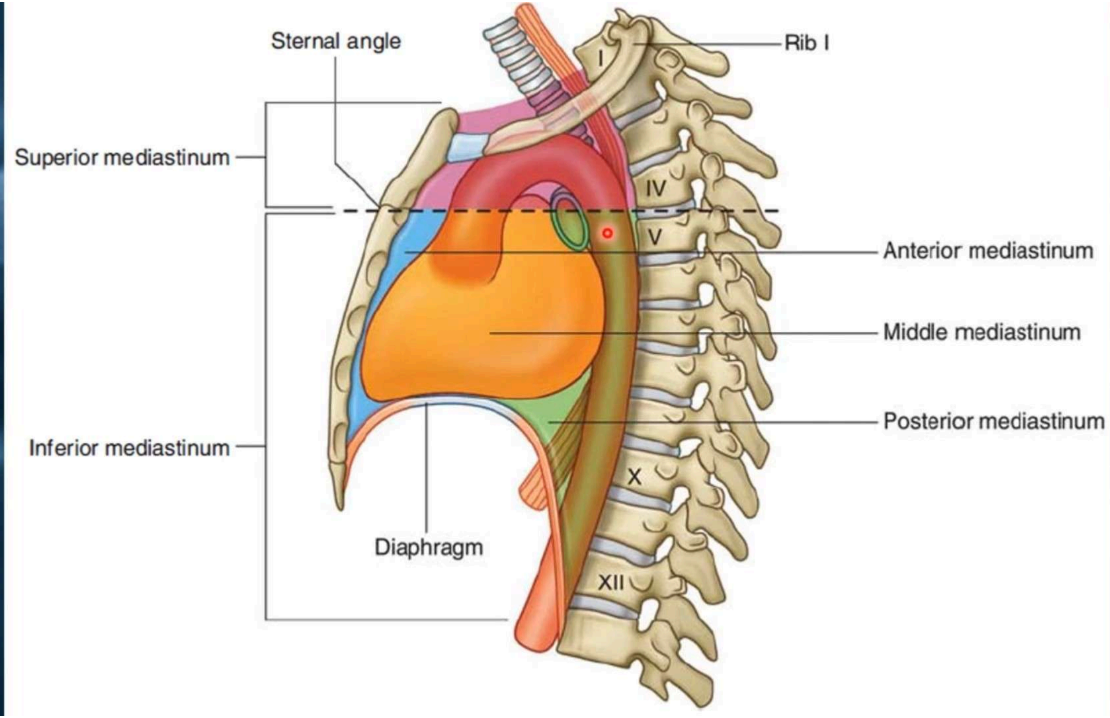
- **12 pairs of ribs** and
- **costal cartilages** that join anteriorly with
- **the sternum and**
- posteriorly with the **thoracic spine vertibra.**

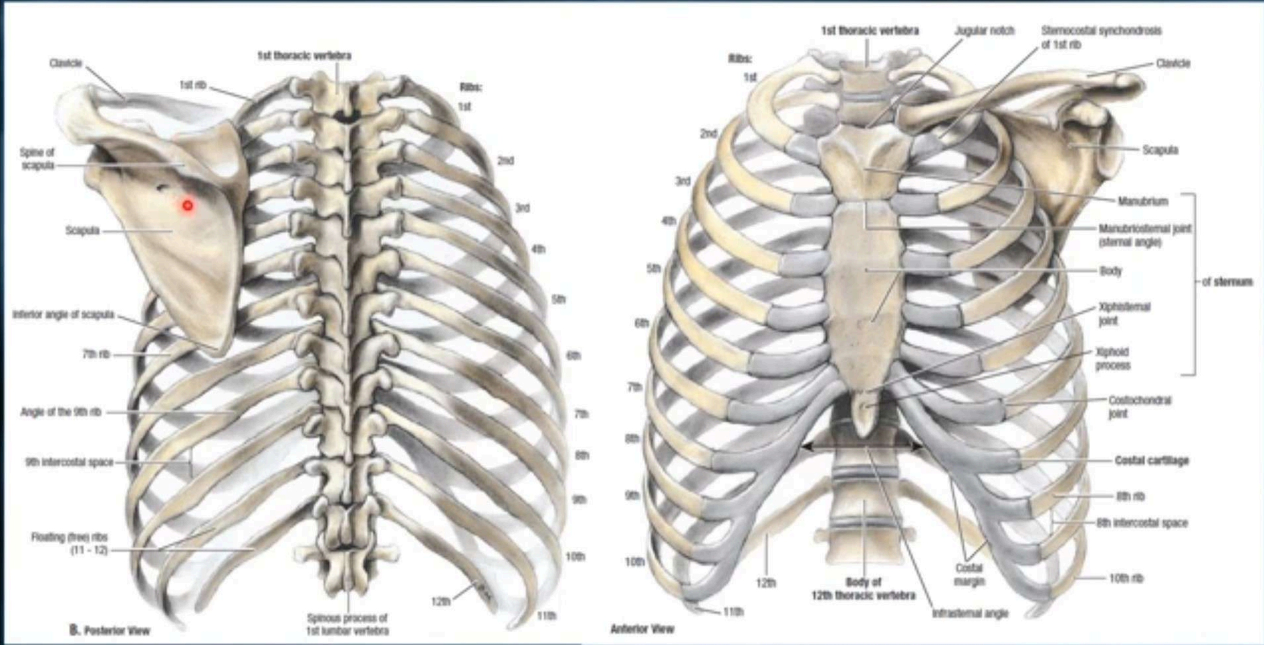


2)Thoracic Cavity:

- It is Lined with a thin layer of tissue **(pleura).**
- **One lung** in each thoracic cavity.
- **Mediastinum** is between the chest cavity
  - Heart, Aorta, Superior and Inferior Vena Cava, Trachea, Major Bronchi, and Esophagus
- **Spinal cord** – protected by vertebral column.



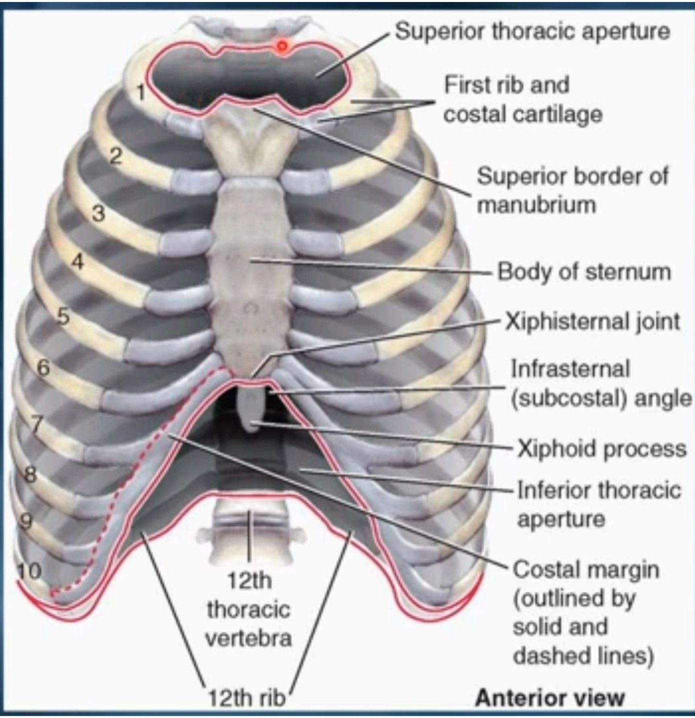

Sternal angle
Rib I
Superior mediastinum
IV
V
Anterior mediastinum
Middle mediastinum
Posterior mediastinum
Inferior mediastinum
X
Diaphragm
XII



Clavicle
1st rib
1st thoracic vertebra
Ribs:
1st
2nd
3rd
4th
5th
6th
7th
8th
9th
10th
11th
12th
Spine of scapula
Scapula
Inferior angle of scapula
7th rib
Angle of the 9th rib
9th intercostal space
Floating (free) ribs (11 - 12)
Spinous process of 1st lumbar vertebra
B. Posterior View
1st thoracic vertebra
Jugular notch
Sternocostal synchondrosis of 1st rib
Clavicle
Scapula
Manubrium
Manubriosternal joint (sternal angle)
Body
of sternum
Xiphisternal joint
Xiphoid process
Costochondral joint
Costal cartilage
8th rib
8th intercostal space
10th rib
Costal margin
Infrasternal angle
Body of 12th thoracic vertebra
12th
11th
10th
Anterior View

The superior thoracic aperture consists of:

- The **body of vertebra TI** posteriorly.
- The **medial margin of rib I** on each side.
- The **superior margin of the manubrium** anteriorly.
- The thoracic cavity is roofed in above the lung apices by **the suprapleural membrane.**



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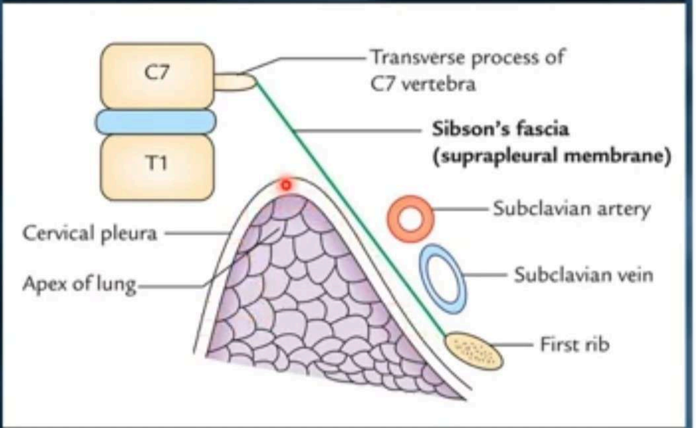
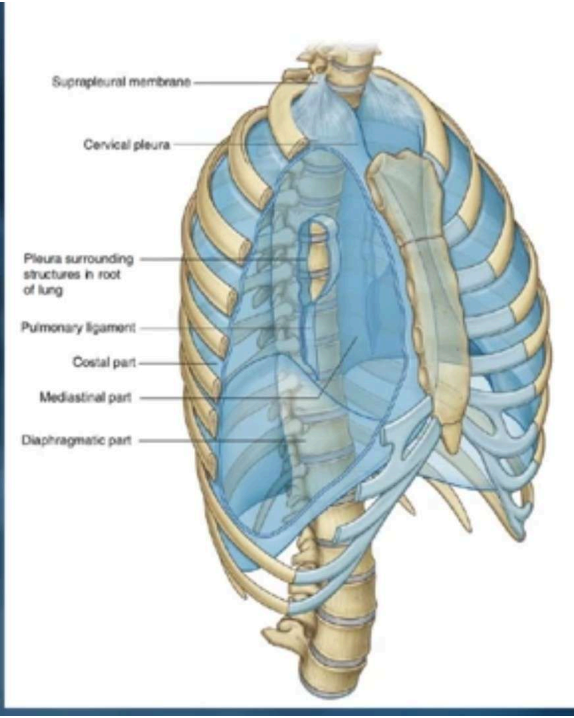
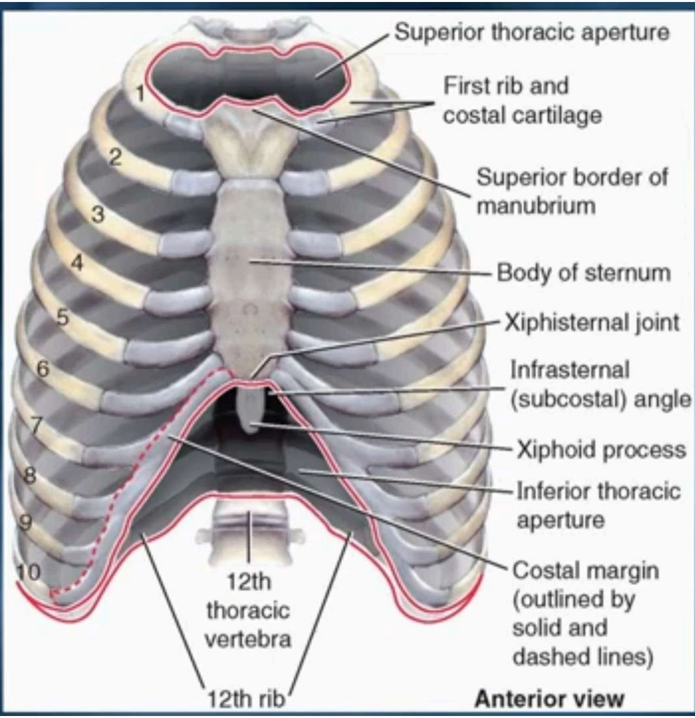
The suprapleural membrane





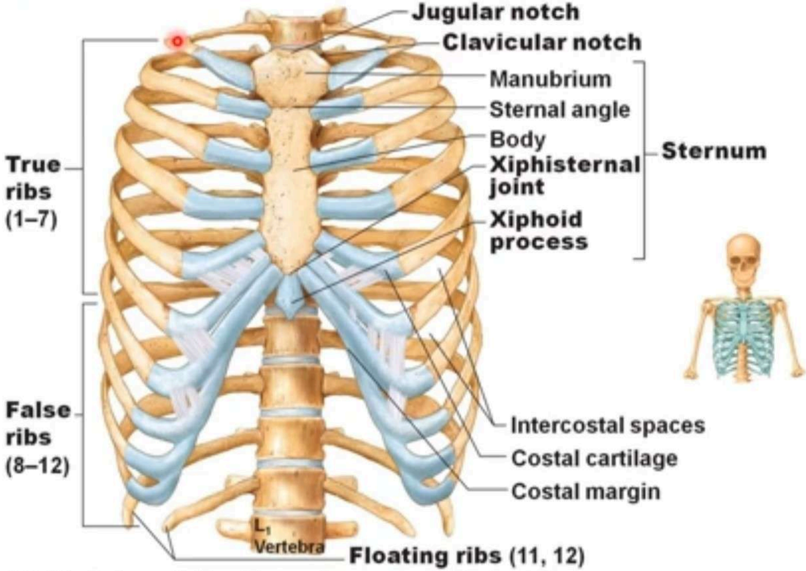
The inferior thoracic aperture consists of

- The inferior thoracic aperture is closed by **the diaphragm**.
- The **structures passing between the abdomen and thorax** pierce or pass posteriorly to the diaphragm.
- **Skeletal elements of the inferior thoracic aperture are consist of:**
- The **body of vertebra TXII** posteriorly.
- The **rib XII** and **the distal end of rib XI** posterolaterally.
- The distal cartilaginous ends of ribs VII to X, which unite to form **the costal margin** anterolaterally.
- The **xiphoid process** anteriorly.



The bone Structure of the Thoracic Cage

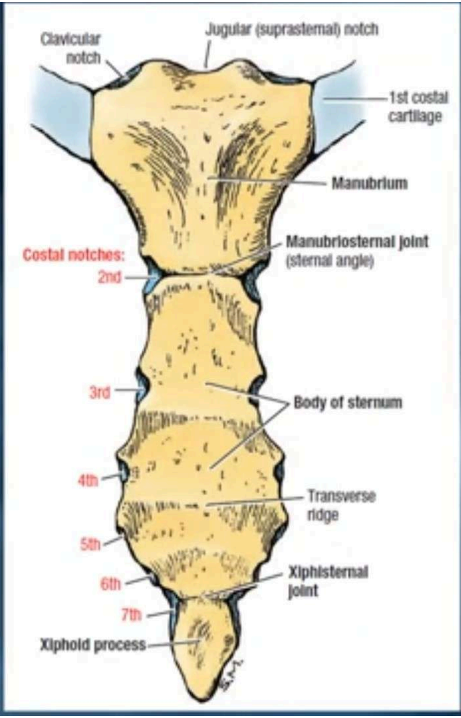
- **Twelve thoracic vertebrae.**
- Twelve **pairs of ribs** and their **costal cartilages.**
- **Sternum.**



(a) Skeleton of the thoracic cage, anterior view

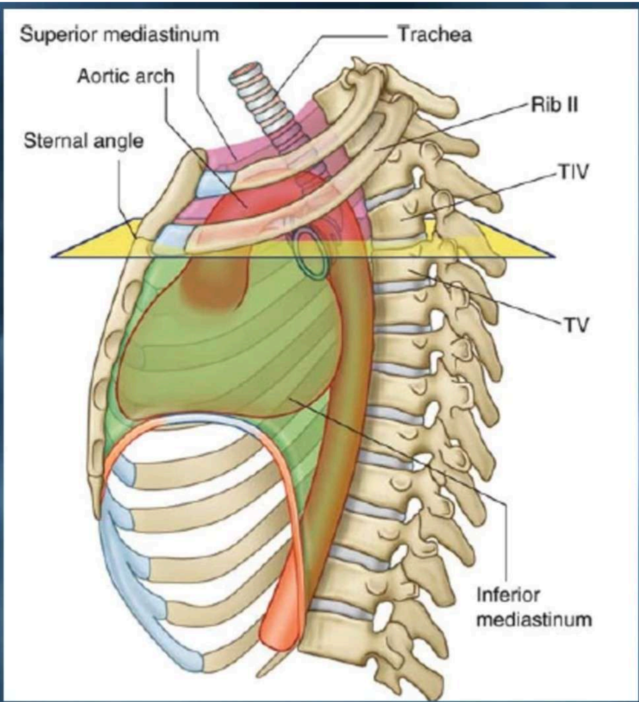
Sternum

- **Manubrium.**
- **Body of sternum:**
  - Articulates with **the second** to **seventh costal cartilages**.
  - articulates with the xiphoid process at the **xiphisternal joint**, which is at level with **the ninth thoracic vertebra.**
- **Xiphoid process (**Xiphisternal synchondrosis):
  - The xiphoid process usually fuses with the body of the sternum during middle age.
- **Joints of the Sternum**
  - The **manubriosternal joint** is a **Secondary cartilaginous joint** (symphysis) between the manubrium and the body of the sternum.
  - The **xiphisternal joint** is a **Primary cartilaginous joint (synchondrosis)** joint between the xiphoid process (cartilage) and the body of the sternum.



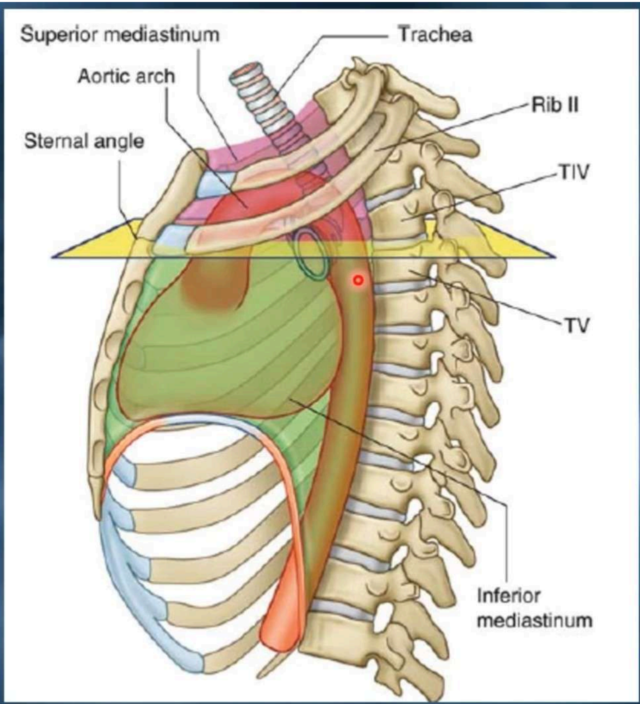
The sternal angle or angle of Louis anatomical landmark for a number of features:

1) Passage of the thoracic duct from right to left behind esophagus.
2) Tracheal Bifurcation.
3) End of the azygos system into SVC.
4) Ligamentum arteriosum.
5) Loop of left recurrent laryngeal nerve around aortic arch.



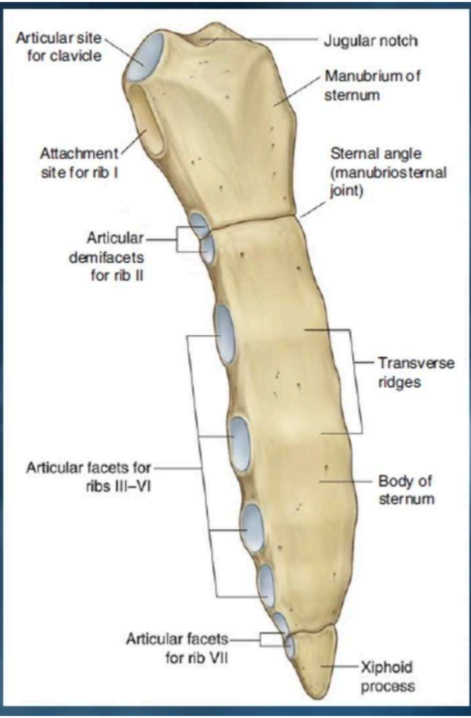
The sternal angle or angle of Louis anatomical landmark for a number of features:

6) Aortic arch starts and ends.
7) Rib Counting: The sternal angle marks the location of the second rib, which is a starting point for counting ribs in physical exams and imaging studies.
8) The horizontal plane between sternal angle and as the T4 and T5 vertebrae in the posterior thoracic spine used as a dividing line between the superior and inferior mediastinum, which helps in describing the anatomical structures within the thoracic cavity.



Clinical note:

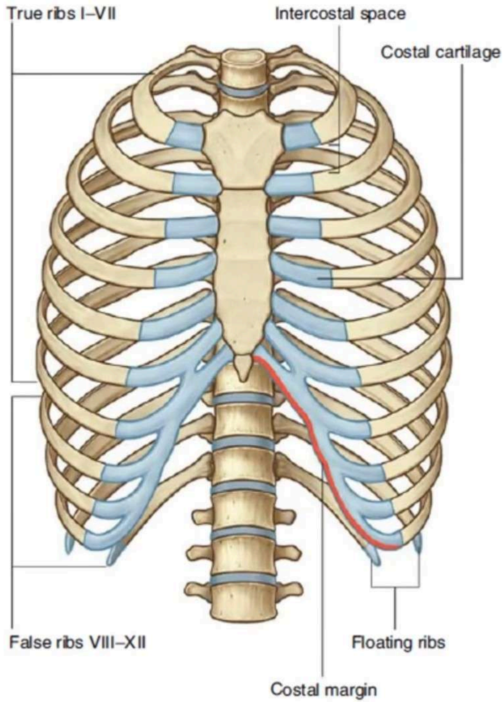
- The sternum possesses red hematopoietic marrow throughout life, it is a common site for marrow biopsy.
- Split in the median plane (median sternotomy) to allow the surgeon to gain easy access to the lungs, heart, and great vessels.



Ribs

- Costa = **Rib bone + its cartilage**
- Vertebrosternal ribs (true):
  1-7
- Vertebrochondral ribs (false):
  8-10
- Vertebral ribs (floating):
  11-12

➢ **Typical and Atypical ribs**



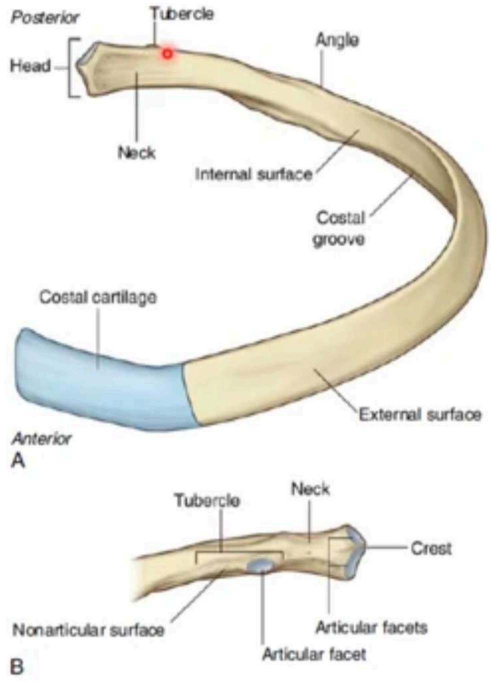
Typical Rib

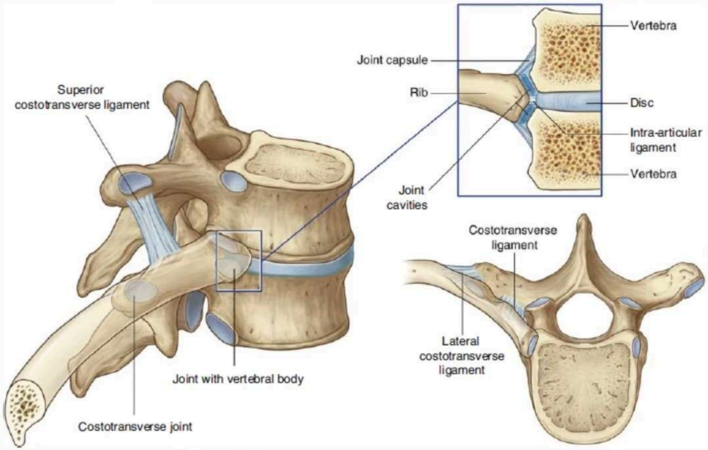
- **Head:**

  Two articular facets for articulation with costal demifacets on adjacent thoracic vertebrae.
- **Neck:**

  Narrow portion between head and tubercle.
- **Tubercle:**

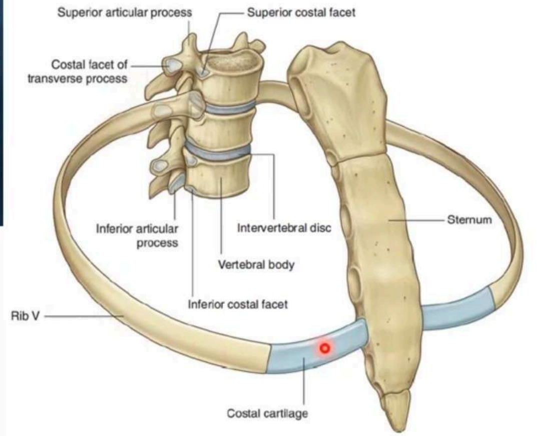
  Articulates with transverse process of vertebra with same number.
- **Shaft.**
- **Sternal extremity.**
- **Angle.**

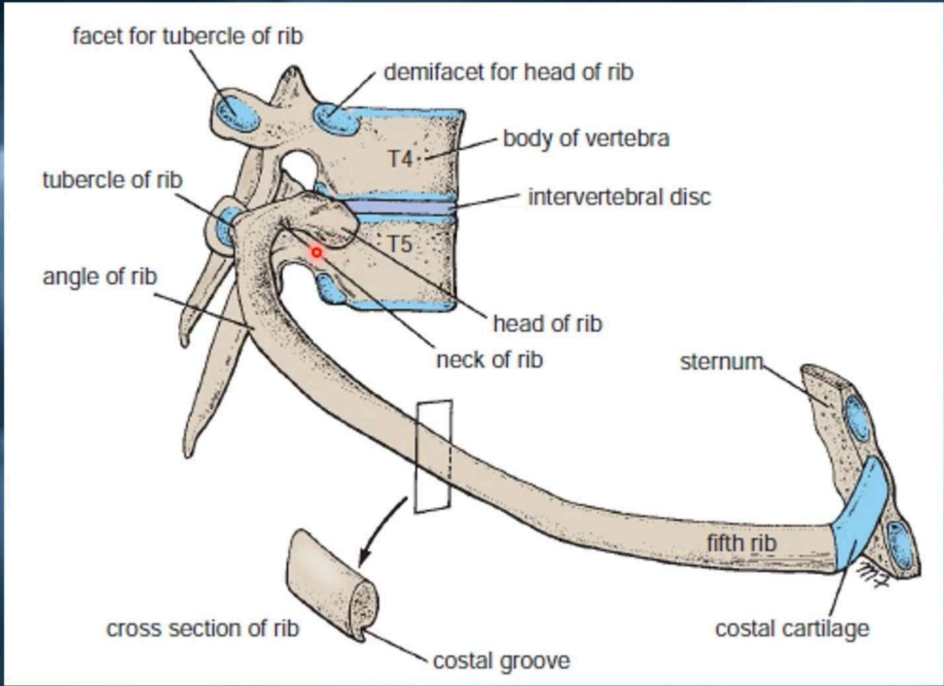




Superolateral view

Superior view




facet for tubercle of rib
demifacet for head of rib
body of vertebra
T4
intervertebral disc
tubercle of rib
T5
angle of rib
head of rib
neck of rib
sternum
fifth rib
cross section of rib
costal cartilage
costal groove

Typical Rib

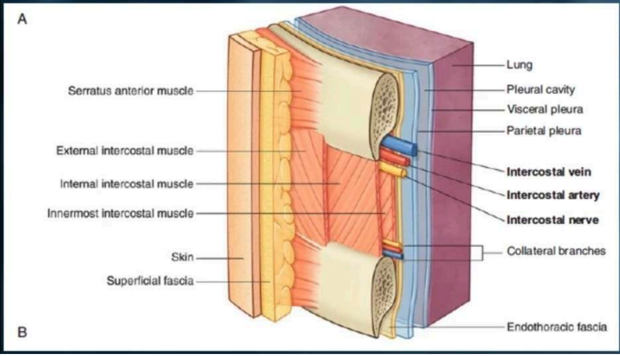
- **Costal groove:**

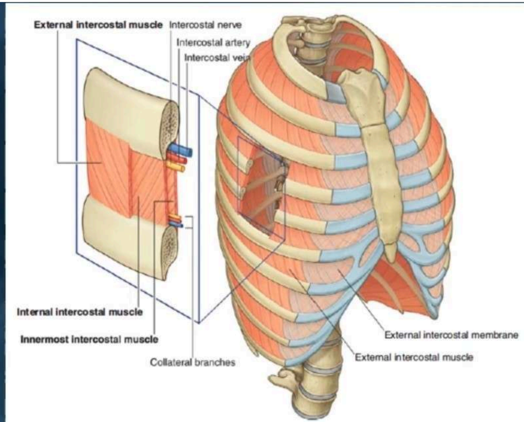
  Contains superior to inferior:

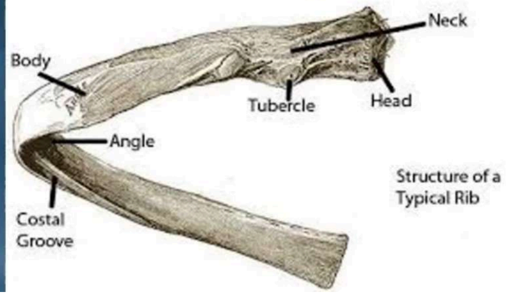
  Intercostal vein

  Intercostal artery

  Intercostal nerve

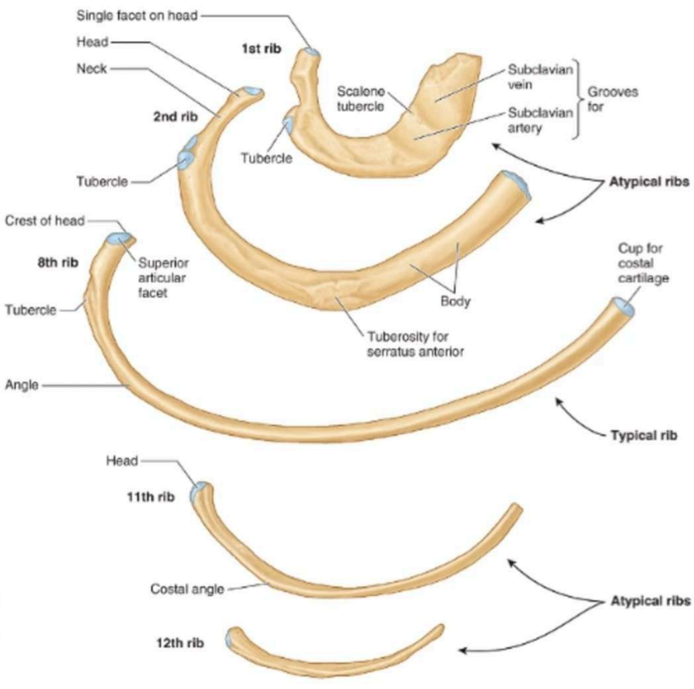






Atypical Ribs

- **1st , 2nd ,10th , 11th** and **12th** pair of ribs are atypical ribs:
- In the posteriour end of atypical ribs, they have single fascet.
- There is **no costral groove** in it.
- There is **no any angle**.
- The **1st rib** forms **primary cartilegenous joint** with manubrum.



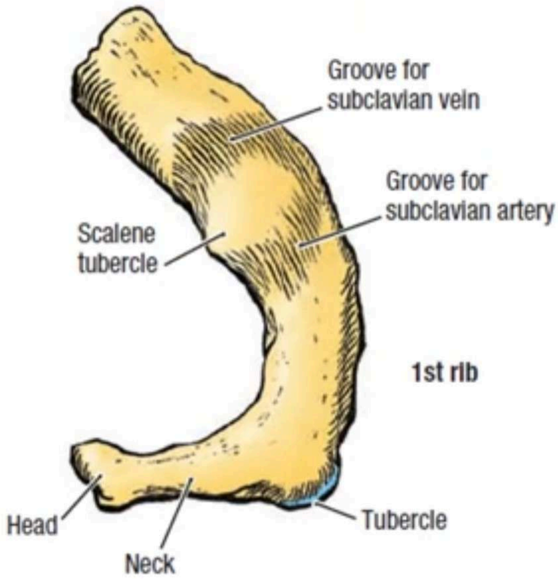
First Rib

- Flattened in a horizontal (transverse) plane.
- **Scalene tubercle**:

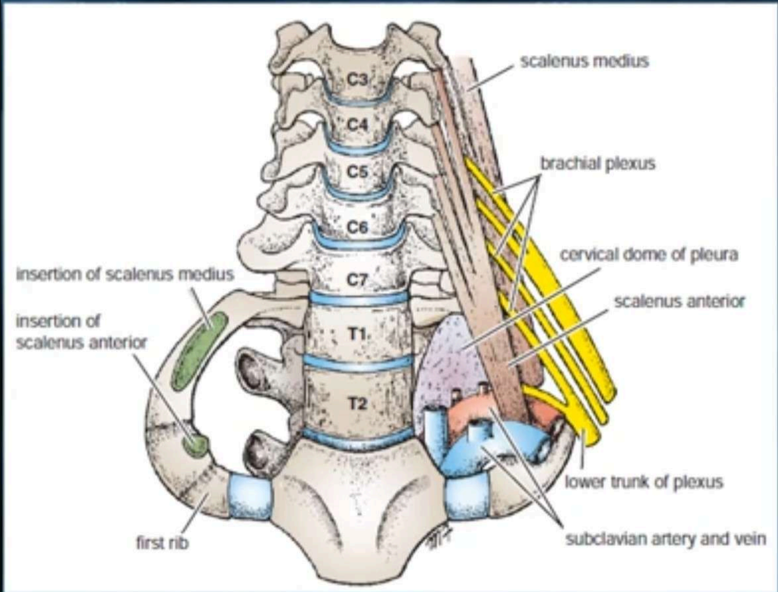
  For insertion of scalenus anterior muscle.
- **Shallow groove for subclavian vein:**

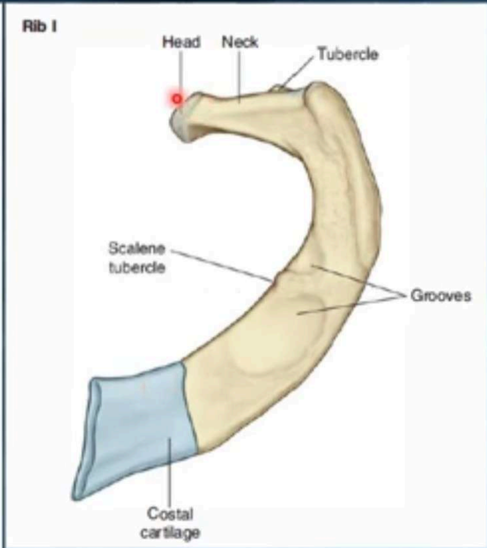
  Anterior to tubercle.
- **Shallow groove for subclavian artery:**

  Posterior to tubercle.



Superior View


scalenus medius
C3
C4
C5
C6
C7
T1
T2
brachial plexus
cervical dome of pleura
insertion of scalenus medius
scalenus anterior
insertion of scalenus anterior
lower trunk of plexus
first rib
subclavian artery and vein


Rib I
Head
Neck
Tubercle
Scalene tubercle
Grooves
Costal cartilage

2) Second Rib:

- Has two articular facets on its head, which articulate with the bodies of the first and second thoracic vertebrae.
- It is a rough area on its upper surface, the tuberosity for serratus anterior muscle originates.
- Is about twice as long as the first rib.

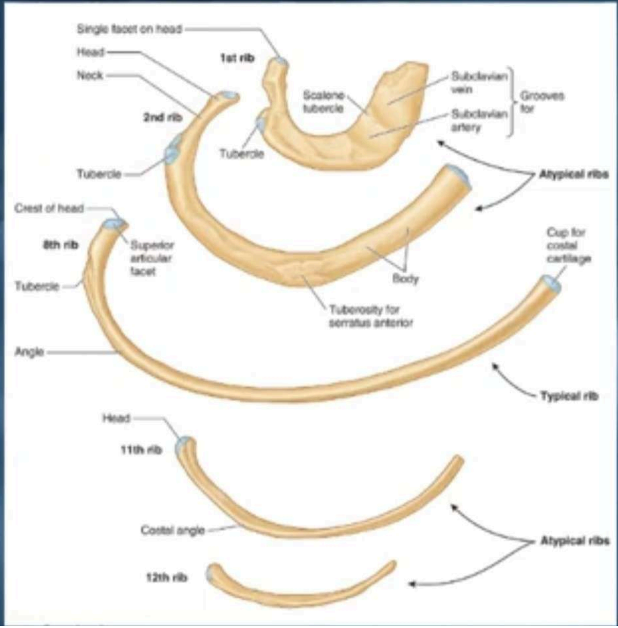
3) Tenth Rib:

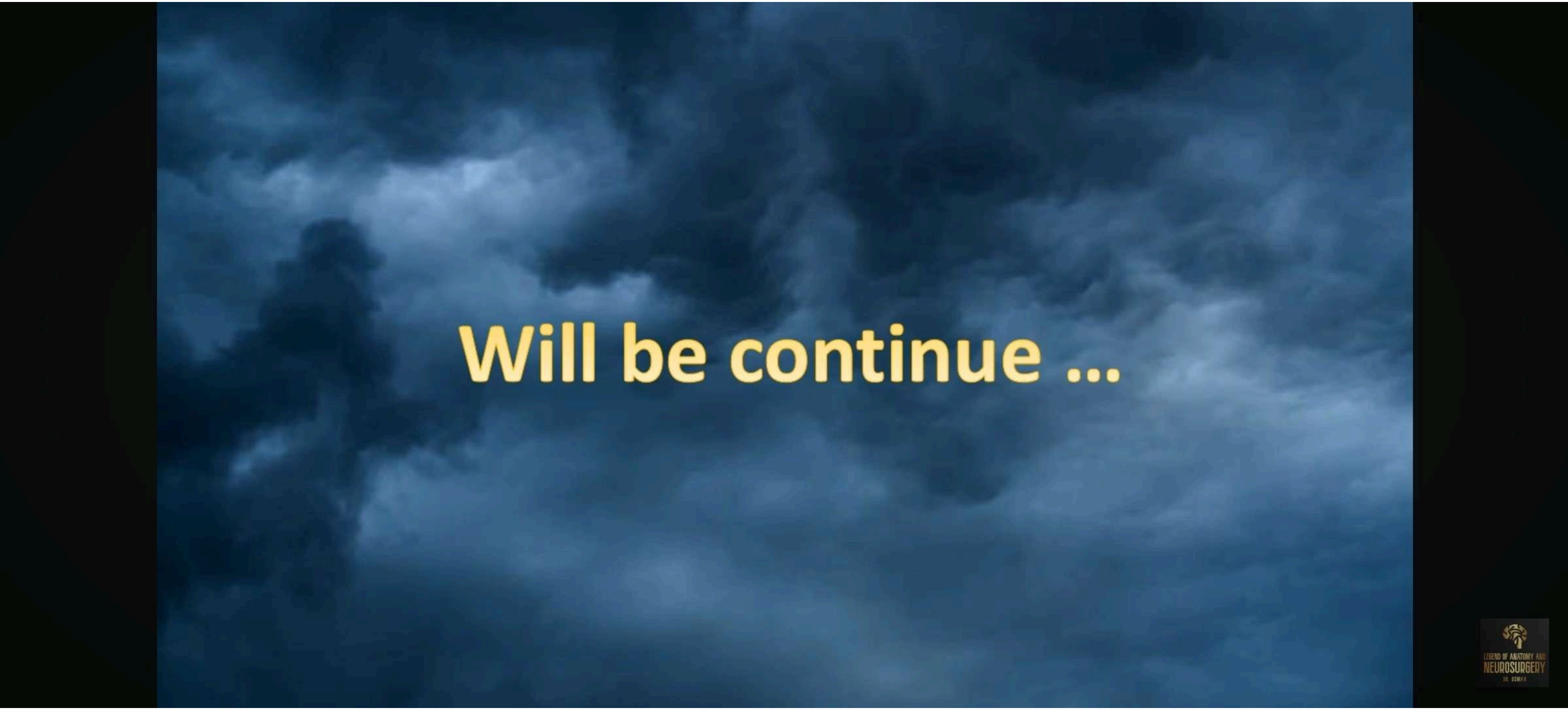
- Has a single articular facet on its head, which articulates with the 10th thoracic vertebra.

4) Eleventh and Twelfth Ribs

- Have a single articular facet on their heads.
- Have no neck or tubercle.

Atypical Ribs

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